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ut the socioeconomic profile of respondents: verage monthly family income, place of birth

one cultivar (Pirâmide Ornamental) (Fig. 1).

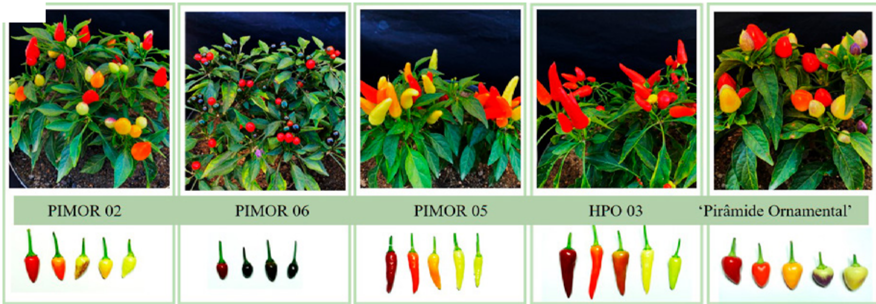

Fig. 1. Pre-cultivars and commercial control of ornamental peppers (*Capsicum annuum* L.).

All pre-cultivars belong to the botanical variety *C. annuum* var. *annuum*, except PIMOR 05 which belongs to *C. annuum* var. *glabriusculum*. The preference regarding isolated attributes of ornamental peppers was also evaluated (flower, fruit and pungency), as well as in relation to the size of the pots. The form was illustrated with images of flowers, fruits and frontal views of the plants, in three different pot volumes (0.75, 2 and 5 dm^3), of all evaluated genotypes.

The link to the virtual questionnaire was shared by the authors in their networks of contacts via social media (Facebook and Instagram and WhatsApp), by email to all students and professors of the Postgraduate Program in Genetics and Plant Breeding/UENF and on official UENF websites and profiles. The online survey took place over two weeks, from July 29 to August 11, 2020. The data resulting from the consumer preference survey were analyzed using descriptive statistics. The Chi-Square (χ^2) association test between crop options and preferences and the socioeconomic conditions of the respondents was carried out in the Genes program (Cruz, 2016).

that grew spontaneously in the backyards of the houses was reported by 6.0% of the respondents, especially by residents of the North region. In the South and Northeast regions, the formal market was the main supplier of seeds for cultivation.

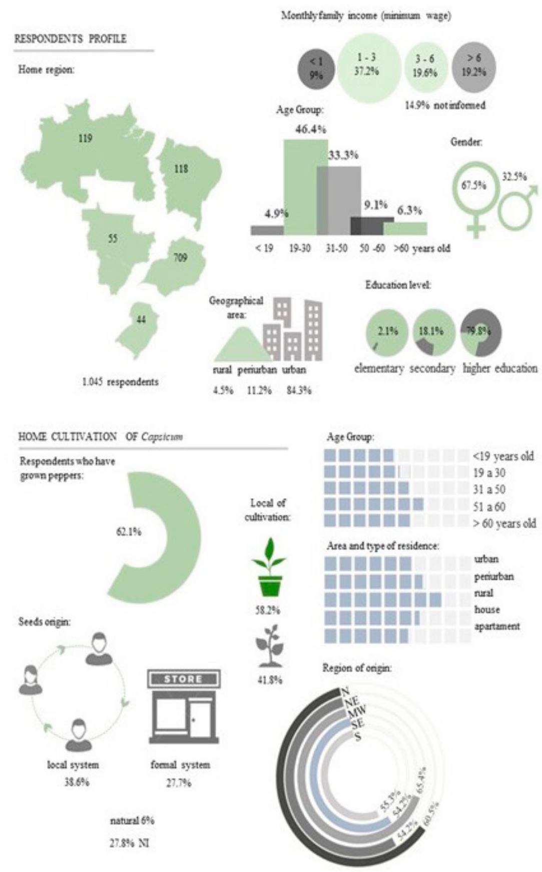

Fig. 2. Illustrated results of the opinion poll on the cultivation and management of species of the genus *Capsicum*, in Brazil.

The Chi-Square test detected a significant association between almost all socioeconomic variables in relation to the home cultivation and management of peppers, except for gender and income. The home region of the respondents was significantly associated with the home cultivation practice (χ^2 = 13.54; p = 0.009) and with the way peppers were cultivated (potted or soil) (χ^2 = 19.89; p = 0.005). North and Central-West were the er 60 years old. The level of education was with the type of cultivation (χ^2 = 43.5; $p \leq$ n home gardens or orchards predominated d only completed elementary school and incomplete high school.

Residence area influenced the cultivation and pepper practices (χ^2 = 10.2; p = 0.006), the type of cultivation (χ^2 = 25.7; $p \leq 0.01$) and the way of acquiring seeds (χ^2= 40.2; $p \leq 0.001$). Respondents who declare that they have already grown pepper, mostly reside in rural areas, an area in which the acquisition of seeds through local networks was more common. In urban areas, potted cultivation was more prominent. The type of property was also positively associated with the type of cultivation (χ^2 = 20.3; $p \leq 0.01$). Potted cultivation was more frequent among those who live in apartments (72.2%), compared to those who live in houses (51.3%).

In relation to ornamental peppers, most respondents (89.8%) revealed that they were interested in purchasing or had already purchased a pepper plant for this purpose. High acceptance (>80%) was common for both genders, regions of Brazil and assessed income levels. Participants' preferences regarding pot sizes, genotypes and their attributes (flower color, fruit type) were investigated based on real images. The preference for 2 dm^3 pots was the majority among respondents at almost all levels of socioeconomic variables. The final ranking for preferred pot types was: 2 (59.3%), 5 (30.2%) and 0.75 dm^3 (10.4%) (Fig. 3).

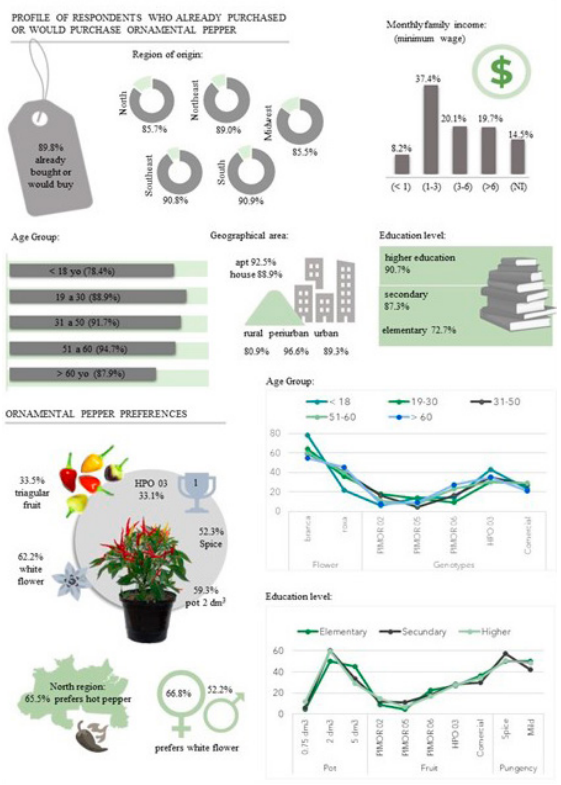

Fig. 3. Illustrated results of the study of consumer acceptance and preferences regarding the ornamental attributes of four pre-cultivars of *Capsicum annuum* L.

White flowers were preferred over purple ones (62.2%), regardless of age, income, education level, area or region of residence of the respondents. Regarding the respondents' birthplace, the Northeast was the only region in which a balanced preference was observed in this regard. Half of the respondents preferred white flowers and the other half purple flowers. The fruits of 'Pirâmide Ornamental' (triangular shape and five verified for the PIMOR 05 genotype (7.7%) (almost round shape and three different colors, along the maturation stages). When the overall phenotype of the genotypes was evaluated, HPO 03 had the highest preference (33.1%). 'Pirâmide Ornamental'' took second place in the overall ranking (27.7%), followed by PIMOR 02 (15.5%) and PIMOR 06 (14.0%). The PIMOR 05 genotype remained in the last position (9.7%).

There was a significant association between preference for flower color and gender (χ^2 = 4.4; p = 0.035), with the highest frequencies of preference for white flowers observed among women. The magnitude of acceptance, in relation to ornamental peppers, was higher in the Southeast and South regions. However, home region was only significantly associated with pungency (χ^2 = 12.0; $p \leq 0.01$). The greatest interest in pungent peppers was verified in the North region.

The age group significantly influenced the acceptance in relation to ornamental peppers (χ^2 = 14.4; p = 0.006), and regarding the attributes flower color (χ^2 = 14.4; p = 0.006) and genotype preference (χ^2 = 33.0; p = 0.007). The lowest frequency of ornamental pepper consumers included young people under the age of 18 (78.4%). The highest frequency of consumers was among adults aged 51 to 60 years (94.7%). The young showed a greater preference for white flowers. As for the genotypes, the greater acceptance of HPO 03 was observed among children under 18 years of age and elderly people over 60 years of age. PIMOR 06 was the second genotype preferred by this last group.

There was a significant association between the level of schooling and acceptance of ornamental peppers (χ^2 = 28.5; $p \leq 0.001$) and preferences for pot volume (χ^2 = 25.7; $p \leq 0.001$), type of fruit (χ^2 = 71.5; $p \leq 0.001$) and pungency (χ^2 = 19.0; p = 0.002). The greatest preference for a large pot (5 dm^3) was observed among those who attended incomplete elementary school. On the other hand, the lowest acceptance occurred among those with higher education. Those who bought and/or would buy ornamental peppers were characterized by having a higher level of education. The preference for fruits of the HPO 03 genotype predominated among respondents with complete elementary and high school levels and incomplete higher education. In these cases, it was observed that 63.6% of respondents with complete elementary education preferred hot peppers to sweet ones. Residence area significantly influenced the acceptance of ornamental peppers (χ^2 = 12.6; p = 0.002), and the highest acceptance was observed among residents of urban and peri-urban areas.

Although the acceptance of ornamental peppers was higher among respondents who live in apartments and receive three minimum wages or more, both the type of property and income did not significantly influence interest and preferences in relation to these plants.

Discussion

This research included respondents from all Brazilian states, from different generations, income levels and education. However, because the authors' contact networks were one of the main means of disseminating the online questionnaire, there was a predominance of respondents with a profile close to the authors. This bias is because the authors' contact networks were one of the main means of disseminating the online questionnaire. It is a predicted trend in this type of study and tends to be reduced over the shares and by the number of respondents (Ball et al., 2019).

The predominance of respondents who stated that they cultivate peppers (*Capsicum* spp.) at home, even in urban areas, reinforces the great popularity of this crop among Brazilians. Peppers are common in home gardens across the country (Ranieri and Zanirato, 2021), certainly, because they are part of several regional culinary traditions. Its use is a cultural heritage of indigenous peoples (Nascimento Filho et al., 2007). Heiser and Smith (1953) mention that naturalist and geographer Alexander Humboldt during his travels to Brazil, between 1799 and 1804, observed and wrote that peppers were as essential for native peoples as salt was for whites.

The respondents' area of residence and age group were the socioeconomic variables that most influenced the habit of growing peppers and other aspects of cultivation, such as how to obtain seeds or seedlings.

ocal level. In the South region, where the lidated and there is a greater tradition of plants (Junqueira and Peetz, 2017), the

The management of spontaneously growing peppers was more frequent among respondents from the North region. This region is considered a center of diversity for the genus *Capsicum* (Alves et al., 2022). In the state of Roraima, a wild pepper that occurs spontaneously in mountainous regions was cataloged (Nascimento Filho et al., 2007). Some people consider spontaneously growing peppers to be healthier and more productive (Egerer et al., 2018). The indigenous people believe that this pepper is planted in their gardens by "Curupira", Fig. of Brazilian folklore, considered "guardian of the forest". Studies show that the biggest disperser of the genus *Capsicum* are birds, which do not have receptors that detect capsaicinoids, responsible for pungency (Egerer et al. 2018). On an Island Micronesia there was a decline in donne' sali pepper (*C. frutescens*), due to the extinction of almost all frugivores, preyed upon by the invasive snake *Boiga irregularis* (Egerer et al., 2018).

It was observed that growing peppers in pots is quite popular in Brazil. Especially among residents of urban areas. This habit possibly favors the acceptance of ornamental peppers. The consumption of potted plants is a global trend (Darras, 2020; Salachna, 2022), followed by Brazilians. A survey that analyzed the behavior of consumers of flowers and ornamental plants across the country identified that potted plants are the most purchased products in this sector (Paiva et al., 2020). Socioeconomic conditions may be related to this trend. Potted plants are better suited to periods of economic recession and lack of time and space (Yano and Amorim, 2022), and enable indoor cultivation, intensified during the social isolation imposed by the COVID-19 pandemic (Reis et al., 2020, Búlgari et al., 2021), popularized through social media.

Paiva et al. (2020) observed that women, with incomes above 4 minimum wages, were the largest consumers of ornamental plants. In the present research, on the other hand, gender and income did not influence the acceptance of ornamental pepper. As already mentioned above, the fact that peppers are a widely used condiment in Brazilian cuisine possibly contributes to its general acceptance, among men and women of all social classes.

Considering the variables that influenced the acceptance of ornamental pepper, it was observed that the profile of potential consumers concentrated on adults, residents of peri-urban and urban areas with a high level of education. Adults aged 30 to 70 were identified as the main consumers of ornamental plants in the European market (Gabellini and Scaramuzzi, 2022).

Among the three volumes of pots submitted to the preference survey, the medium one, followed by the large one, were preferred. Although small pots are more practical for domestic handling, take up less space and require less substrate and fertilizers, they limit the development of plants, making the pot-plant set less attractive.

Regional, generational, gender, and educational differences significantly influenced the preference regarding different attributes of flowers and fruits of the pre-cultivars of ornamental pepper evaluated. This indicates that the genetic variability identified in the pre-cultivars has the potential to be explored in the development of new ornamental pepper cultivars, meeting the different preferences of consumers. There are different market niches that can be explored and that should not be ignored. The diversity of products available increases interest in different species in the ornamental segment (Salachna, 2022).

The genotypes classified with the highest ornamental value (HPO 03 and 'Pirâmide Ornamental') were also preferred in relation to the type of fruit. As both are characterized by the simultaneous occurrence of fruits of different colors, contrasting with the foliage, this can be considered a relevant ornamental attribute, with a great influence on the preference for genotypes. Neitzke et al. (2016), evaluating the preference for ornamental pepper accessions from *Capsicum*, also observed diversity of shapes and

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sis, writing – original draft. **RBB:** formal & editing. **RR:** conceptualization, project ervision.

Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Data Availability Statement

Data will be made available on request.

HEISER, C.B.; SMITH, P.G. The cultivated *Capsicum* peppers. **Economic Botany**, v.7, n.3, p.214-227, 1953.